

GANDAN HURYEE, Mongolia

WMO: 442210

Lat: 49.7006N

Lon: 96.3633E

Elev: 2181

StdP: 77.72

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.1	-42.1	-46.4	0.0	-43.8	-44.5	0.1	-42.0	6.0	-22.6	4.5	-25.0	0.3	180	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.5	14.0	26.5	13.4	24.8	12.7	16.0	22.8	15.0	22.6	14.1	21.9	2.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	12.8	18.6	12.4	11.7	17.6	11.3	10.9	17.1	53.3	22.9	49.8	22.6	47.0	21.9	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.1	5.2	DB	-45.8	32.2	3.3	1.7	-48.1	33.4	-50.0	34.4	-51.8	35.4	-54.2	36.7	
			WB	-46.1	17.8	2.9	2.0	-48.2	19.3	-49.9	20.4	-51.6	21.5	-53.7	23.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.7	-32.3	-28.0	-15.5	1.0	7.6	14.9	17.3	14.8	7.8	-1.4	-16.2	-27.8	(6)
(7)		DBStd	18.33	5.69	6.50	7.24	6.30	4.50	3.75	3.01	3.44	4.34	5.56	7.23	5.76	(7)
(8)		HDD10.0	5952	1310	1063	791	272	102	6	0	5	93	353	785	1172	(8)
(9)		HDD18.3	8446	1568	1297	1049	521	334	111	54	118	317	611	1035	1431	(9)
(10)		CDD10.0	587	0	0	0	1	26	154	228	153	26	0	0	0	(10)
(11)		CDD18.3	40	0	0	0	0	0	9	23	8	0	0	0	0	(11)
(12)		CDH23.3	708	0	0	0	1	16	184	336	159	13	0	0	0	(12)
(13)		CDH26.7	154	0	0	0	0	1	33	84	36	1	0	0	0	(13)
(14)	Wind	WSAvg	1.5	0.5	0.7	1.3	1.9	2.5	2.2	1.9	1.8	1.7	1.4	0.9	0.6	(14)
(15)	Precipitation	PrecAvg	193	4	2	4	8	10	30	53	43	17	8	6	8	(15)
(16)		PrecMax	269	18	8	13	28	25	56	82	116	51	23	16	41	(16)
(17)		PrecMin	133	0	0	0	0	1	5	15	6	4	0	1	1	(17)
(18)		PrecStd	39	4	2	3	7	6	15	17	23	11	7	5	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-15.2	-7.7	5.6	18.9	26.3	29.6	32.1	30.6	25.1	15.5	3.3	-9.1	(19)
(20)		0.4%	MCWB	-15.9	-9.5	-0.1	6.7	11.0	13.7	17.0	14.0	11.5	5.5	-1.4	-10.3	(20)
(21)		2%	DB	-18.7	-11.9	1.5	16.3	22.7	27.4	29.3	27.0	22.1	12.6	0.1	-13.2	(21)
(22)		2%	MCWB	-19.4	-12.9	-1.5	5.7	9.8	13.3	15.0	13.4	10.3	4.4	-2.7	-13.9	(22)
(23)		5%	DB	-21.1	-15.1	-1.2	13.7	20.3	25.4	27.0	24.8	19.4	10.0	-2.4	-15.8	(23)
(24)		5%	MCWB	-21.6	-15.9	-3.7	4.7	8.6	12.5	14.1	12.8	9.2	2.8	-4.4	-16.3	(24)
(25)		10%	DB	-23.3	-17.8	-3.9	11.1	17.8	23.5	25.0	22.8	16.7	7.5	-5.4	-18.6	(25)
(26)		10%	MCWB	-23.7	-18.5	-5.9	3.8	7.4	11.8	13.3	12.3	7.9	1.8	-6.9	-19.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.0	-9.5	0.7	8.0	12.6	16.0	18.5	16.2	12.8	6.2	-0.6	-10.2	(27)
(28)		0.4%	MCDWB	-15.3	-7.7	4.5	15.7	22.4	23.6	27.9	22.6	21.5	13.9	2.7	-9.1	(28)
(29)		2%	WB	-19.3	-12.9	-1.4	6.7	10.5	14.9	16.7	14.9	11.3	4.9	-2.1	-13.9	(29)
(30)		2%	MCDWB	-18.7	-11.8	1.7	14.1	21.3	23.4	25.1	22.5	19.5	11.6	0.4	-13.3	(30)
(31)		5%	WB	-21.5	-15.9	-3.7	5.4	9.2	13.8	15.6	14.1	10.0	3.6	-3.8	-16.4	(31)
(32)		5%	MCDWB	-21.1	-15.1	-1.3	12.6	19.0	22.0	22.6	21.9	17.9	9.6	-1.6	-16.0	(32)
(33)		10%	WB	-23.7	-18.5	-5.9	4.0	7.9	12.9	14.6	13.3	8.8	2.1	-5.8	-19.1	(33)
(34)		10%	MCDWB	-23.3	-17.9	-3.9	10.9	16.8	21.0	22.1	20.8	15.7	7.7	-4.2	-18.6	(34)
(35)	Mean Daily Temperature Range		MDBR	9.8	12.1	14.3	12.3	14.1	13.5	12.3	13.2	14.4	13.4	10.7	9.3	(35)
(36)		5% DB	MCDBR	11.1	13.7	14.4	16.2	19.0	16.8	16.6	17.3	18.9	17.5	13.1	10.6	(36)
(37)			MCWBR	10.7	12.7	11.4	7.4	9.3	6.5	6.2	6.8	8.9	10.1	8.4	10.0	(37)
(38)			MCDBR	11.2	13.6	13.7	14.3	17.0	13.2	12.7	13.6	16.3	16.5	10.4	10.7	(38)
(39)		5% WB	MCWBR	10.8	12.6	11.0	7.2	8.8	5.7	5.4	5.7	7.9	9.8	8.2	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.203	0.224	0.256	0.291	0.299	0.301	0.305	0.290	0.260	0.245	0.226	0.207	(40)	
(41)		taud	2.355	2.319	2.315	2.416	2.428	2.507	2.517	2.540	2.583	2.546	2.435	2.344	(41)	
(42)		Ebn at Noon	899	951	965	960	960	957	948	951	958	921	858	838	(42)	
(43)		Edh at Noon	62	84	102	105	109	101	99	92	78	67	57	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.56	2.59	4.17	5.54	6.39	6.56	6.11	5.32	4.33	2.84	1.59	1.20	(44)	
(45)		RadStd	0.07	0.12	0.14	0.24	0.32	0.26	0.30	0.26	0.27	0.17	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-2.41	N/A	-0.88	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page